Why the preference for Pixel Experience?

After a lot of research, I came to the conclusion that Pixel Experience is best suited to general user requirements, so I decided to go with it. As the name suggests, it is supposed to give you a Google Pixel like experience on your device. This ROM comes with preloaded Google apps so there's no need to flash externally. Over the air (OTA) updates are provided by the community regularly. I have used this ROM for the past six months, and am getting monthly security patch updates with other bug fixes and enhancements.

Pixel Experience is a lightweight and less customisable ROM, so it consumes less battery. The battery performance is outstanding and beyond expectations.

Are there security concerns when a custom ROM is installed?

It is not true that the installation of a custom ROM compromises the security of a phone or device. Behind every custom ROM there is a large community and thousands of users who test it.

For custom ROM installation, you don't need to root your device — it is 100 per cent safe and secure. If we keep this discussion specific to the Pixel Experience ROM, then it is a pure vanilla Android ROM developed for Nexus and Pixel devices, and ported by developers and maintainers to specific Android devices.

Before installing a custom ROM, you need to unlock the bootloader. During the bootloader unlock process, you will see a warning from the vendor that states your phone will be less secure after unlocking the bootloader. The reason for this is that an unlocked phone can be used to install a fresh ROM without any permission from device manufacturers or the owner of the device. So, stolen or lost devices can be reused by flashing a ROM. Anyway, there are a number of methods that can be used to unlock the bootloader unofficially and install the ROM without permission from the manufacturer.

Custom ROMs are more secure than stock ROMs because of the latest updates provided by the community. Device manufacturers are profit making companies. They want their customers to upgrade their phones after two years; so they stop providing support and stop pushing software updates! Custom ROMs, on the other hand, are driven by non-profit communities. They run on community support and donations.

What does it mean to root an Android device, and is this really required before flashing a custom ROM?

On a Windows machine, there is an administrator account which has all the privileges. Similarly, in Linux too, there is the concept of a root account. Android uses the Linux kernel; so all the OS internals are the same as in Linux.

Your Android phone uses Linux permissions and file system ownership. You are a user when you sign in and you can do only certain things based on your user permissions. In all Android devices, the root user is hidden by the vendor to avoid misuse.

Rooting an Android phone is to jail-break the phone to allow the user to dive deep into the device. I personally recommend that you do not root your device, because doing so is really not required to flash a custom ROM to the device.

Here are a few reasons why you should not root your device:

1. Rooting can give your apps complete control of the system, and there is a chance of misuse of power.
2. Google officially does not support rooted devices.
3. Banking applications, BHIM, UPI, Google Pay, PhonePe and Paytm will not work on rooted devices.
4. There is a myth that rooting of a phone is required to flash a custom ROM, but that is not true. You only need to unlock the bootloader to do so.

What is a bootloader and why should you unlock it before flashing a custom ROM?

A bootloader is the proprietary image responsible for bringing up the kernel on a device. It is nothing but a guard for the device, and is responsible for initialising trust between root and user. The bootloader may directly flash the OS to the partition or we can use custom recovery to do the same thing.

In this article we will use Team Win custom recovery to flash the operating system in a device.

In Microsoft Windows terminology, there is the concept of a BIOS which is the same as a bootloader. Let's look at an example. When we install Linux alongside Windows on a laptop or PC, there is bootloader called GRUB which allows the user to boot either Windows or Linux. A bootloader points to the OS partition from the file system. At the press of the power button to start the phone, the bootloader initiates the

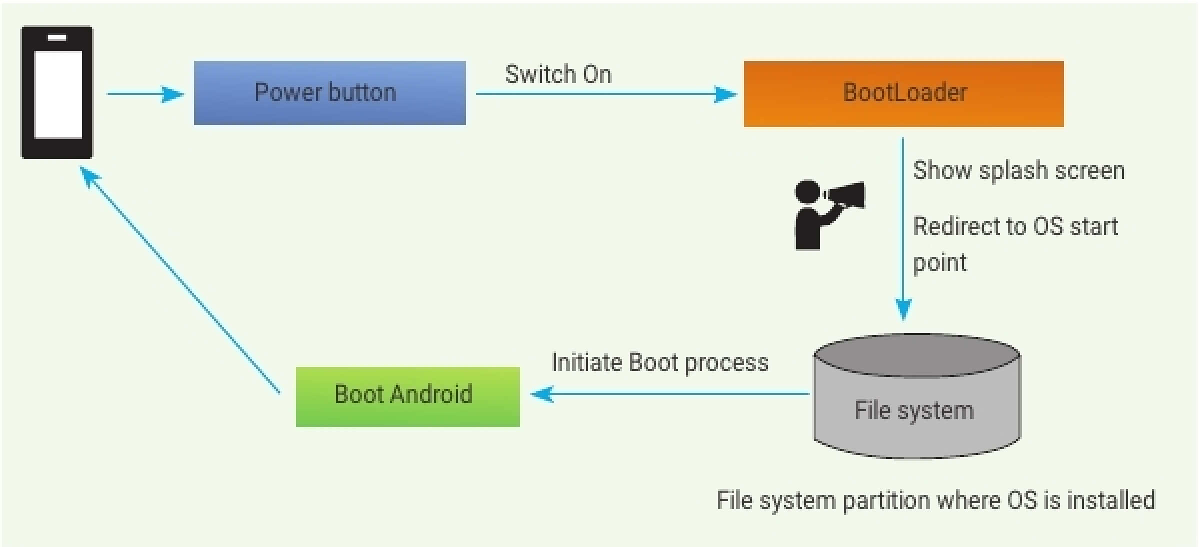

Figure 1: Bootloader